Business Continuity and Disaster Recovery Assignment

GRC205 Incident Management and Business Continuity, Week 21. Northbridge Skills Institute

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Introduction and assumptions

This report responds to the six tasks of the GRC205 assignment. I act as the Business Continuity and Disaster Recovery Coordinator for Northbridge Skills Institute. The main site becomes unavailable at 08:00 WAT on Day 0, a Monday, and stays closed for an estimated 72 hours. Times are in West Africa Time. Where the brief gives no answer I state an assumption. These are listed below as A1 to A9. The results in Task 5 are modelled baseline results and not live tests. They follow the planning steps in NIST SP 800-34 Revision 1 (Swanson et al. 2010).

ID	Assumption
A1	Day 0 is a Monday. Costs are counted from 08:00.
A2	The hourly costs in the brief are separate and do not overlap.
A3	No cyberattack has been confirmed. Compromise is treated as a possibility to check, not a fact.
A4	DB01 incrementals run every hour. So the latest recoverable point is B01 plus eight incrementals (00:00 to 07:00). This is unverified.
A5	The alternate facility is outside the flood zone and has its own power. This is unverified.
A6	Only the two technicians can run restoration actions. Business coordinators cannot.
A7	Staff use their own laptops and phones with Microsoft 365. The 25 alternate workstations are kept for learners.
A8	The Head of Assessment can approve a controlled Microsoft 365 based assessment as an acceptable temporary assessment service.
A9	The offsite provider holds encrypted copies of B01 to B04. Its access independence is not documented, so it is unverified.

Task 1: Business Impact Analysis

1.1 Five-function BIA

The BIA table covers F1 to F5. For each function it records the owner, the people affected, the critical resources, the dependencies, the impact over time, the minimum acceptable service and the priority. The minimum services use Microsoft 365, which stays available through mobile internet, and the alternate facility. Every manual process has an approval and a record control so the temporary work can be checked and reconciled later.

Element	F1 Teaching and assessment	F2 Fees and finance	F3 Admissions and registration	F4 Staff payroll	F5 Records and certificates
Owner role	Head of Assessment	Finance Manager	Registrar	HR and Payroll Manager	Records Officer
Users affected	1,800 learners. 120 sit the 11:00 exam. Teaching staff and invigilators.	Learners paying fees. Finance staff.	Applicants. Registry staff.	90 staff. HR and payroll staff.	Learners and past learners who request records. Records staff.
Critical resources	People: Head of Assessment, 6 invigilators, 2 technicians. Premises: alternate facility (25 workstations from 10:00). Systems: LMS01, DB01. Suppliers: alternate facility, mobile network, Microsoft 365. Information: exam papers, learner lists, accounts.	People: Finance Manager, 2 finance officers. Premises: remote or alternate facility. Systems: FIN01, DB01. Suppliers: payment provider, bank. Information: fee ledger at 07:00, provider transaction reports.	People: Registrar, 2 registry staff. Premises: remote or alternate facility. Systems: REG01, DB01. Suppliers: Microsoft 365 forms, mobile network. Information: applicant list, entry criteria.	People: HR and Payroll Manager, payroll officer, Finance Manager as approver. Premises: remote. Systems: HR01, DB01. Suppliers: bank portal. Information: last approved payroll file, change records.	People: Records Officer. Premises: remote. Systems: FS01. Suppliers: warm site provider. Information: request log, access permissions, certificate templates.
Dependencies	Recovery network, AUTH01, DB01, LMS01. Staff, secure access, communication.	Recovery network, AUTH01, DB01, FIN01. Staff, secure access, communication.	Recovery network, AUTH01, DB01, REG01. Staff, secure access, communication.	Recovery network, AUTH01, DB01, HR01. Staff, secure access, communication.	Recovery network, AUTH01, FS01. Staff, secure access, communication.
Impact over time	0 to 8 h: NGN 480,000, plus NGN 120,000 if no assessment service at 11:00. 8 to 24 h: a further NGN 960,000. 24 to 72 h: teaching backlog, learner stress, integrity risk.	8 h (16:00): fee confirmation due. NGN 320,000. 24 h: NGN 960,000. Reconciliation work grows every hour.	8 h: NGN 160,000. 10 h (18:00): admissions close and applicants may be lost. 24 h: NGN 480,000.	8 h: NGN 80,000. 24 h: NGN 240,000. 28 h (12:00 Day 1): bank deadline. Errors or delay affect 90 staff.	8 h: NGN 40,000. 24 h: NGN 120,000. 48 h: internal service commitment is breached.
Minimum acceptable service	A controlled Microsoft 365 assessment for the 11:00 exam. Normal LMS01 teaching once restored.	Payments continue through the provider. Receipts logged in a controlled sheet. Fees confirmed by 16:00 only for matched payments.	Applications collected by an online form and paper form until 18:00. No admission offers until the Registrar signs off.	No payroll run from temporary data. Use the last approved payroll file with approved changes only.	Urgent requests handled by hand after identity checks. Routine requests wait for FS01.
Approval and record controls	Head of Assessment approves the method. Two invigilators sign each sitting. Submissions exported and logged.	Finance Manager approves. Second person checks each receipt. Numbered receipts and provider transaction IDs.	Registrar approves. Numbered forms. Duplicate check before entry into REG01.	HR Manager and Finance Manager both approve. Duplicate check against the last file.	Records Officer approves. Request log. No release without identity check.
Business priority	1	2	3	4	5

1.2 Impact calculations

The hourly rates are treated as separate costs (A2). With no recovery or continuity measures there is no acceptable assessment service at 11:00. So the one-off cost of NGN 120,000 applies at both 8 hours and 24 hours. I show it under F1. The combined estimate is NGN 1,200,000 at 8 hours and NGN 3,360,000 at 24 hours.

Function	Rate (NGN/h)	8-hour working	8-hour cost (NGN)	24-hour working	24-hour cost (NGN)
F1 Teaching and assessment	60,000	8 × 60,000	480,000	24 × 60,000	1,440,000
F2 Fees and finance	40,000	8 × 40,000	320,000	24 × 40,000	960,000
F3 Admissions and registration	20,000	8 × 20,000	160,000	24 × 20,000	480,000
F4 Staff payroll	10,000	8 × 10,000	80,000	24 × 10,000	240,000
F5 Records and certificates	5,000	8 × 5,000	40,000	24 × 5,000	120,000
Hourly subtotal	135,000	8 × 135,000	1,080,000	24 × 135,000	3,240,000
One-off examination rescheduling (F1)	n/a	Trigger met at 11:00	120,000	Trigger met at 11:00	120,000
Total		1,080,000 + 120,000	**1,200,000**	3,240,000 + 120,000	**3,360,000**

These figures have two main limits. First, a fixed hourly rate assumes harm builds evenly. In practice harm jumps when a deadline is missed, such as fee confirmation at 16:00 or the payroll bank cut-off at 12:00 on Day 1. Second, the figures cover only 24 hours and leave out response costs such as overtime and the alternate facility. They are exercise estimates and not observed losses.

Money is also only one measure of impact. Learner welfare matters when a learner misses an examination. Reputation can fall if fees or results are wrong. Data integrity matters most for payroll and records. A wrong payment or an altered record can harm people long after the outage and cannot be priced from an hourly rate.

1.3 Priorities and dependency risks

The 11:00 examination makes F1 the first business priority. The other deadlines follow in time order. Fee confirmation for F2 is due at 16:00, admissions for F3 close at 18:00 and the payroll bank file for F4 is due at 12:00 on Day 1. F5 has a 48 hour commitment and ranks last. The business order is F1, F2, F3, F4 and F5.

The technical order is different because F1 to F4 share DB01. The network, AUTH01 and the security and backup checks come first. DB01 is restored next. Only then can the four applications start. FS01 needs only AUTH01 and the checks. Although F5 ranks last, I restore FS01 while the first technician restores DB01. This uses an idle technician and delays nothing.

Business priority	Technical restoration order
1 F1 Teaching and assessment	1 Recovery network
2 F2 Fees and finance	2 AUTH01
3 F3 Admissions and registration	3 Security and backup checks
4 F4 Staff payroll	4 DB01 (with FS01 in parallel on the second technician)
5 F5 Records and certificates	5 LMS01, FIN01, HR01 and REG01

Four dependency risks matter most. First, AUTH01 uses a separate local directory and its latest snapshot is eight hours old. If it fails, all five functions stop. Second, DB01 serves four functions, so one broken backup in the chain blocks them all. Third, backup administration shares an administrator identity with production. A mistake or attacker that reaches production may also reach the backups. Fourth, only two technicians are available. One absence would roughly double the restoration time. The alternate facility is a further risk, since it has only 25 workstations and depends on mobile internet.

Task 2: Recovery objectives and backup design

2.1 Objectives

RTO is the time allowed to restore a function after the disruption starts. RPO is how old the restored data may be, counted back from the moment of disruption. MTD is the longest disruption the business can tolerate before harm becomes unacceptable (Swanson et al. 2010). I set each RTO well inside its MTD so that a buffer remains for checks and delays.

I declare a function recovered only when it meets a stated service level. A restored server that nobody has checked is not recovery. Each function needs a successful test transaction, a data check against the checkpoint time and acceptance by the function owner. The RTOs follow the Warm package estimate of 3 to 5 hours and the drill timings in Task 5.

Function	MTD	RTO	RPO	Recovery is declared when	Buffer (MTD − RTO)	Basis
F1	12 h	4 h	1 h	LMS01 login, content and a test submission pass. Data matches the 07:00 checkpoint. Head of Assessment accepts.	8 h	Learners can sit the exam by the workaround, so full LMS01 can follow.
F2	8 h	4 h	1 h	FIN01 opens the ledger as at 07:00. A test receipt posts. Reconciliation with the provider report has started. Finance Manager accepts.	4 h	Fee confirmation at 16:00 (8 h).
F3	10 h	5 h	1 h	REG01 saves a test application. Interim applications are loaded. Registrar accepts.	5 h	Admissions close at 18:00 (10 h).
F4	24 h	6 h	4 h	HR01 opens the last approved payroll. Changes since 07:00 are checked. HR Manager accepts.	18 h	Bank file due 12:00 Day 1 (28 h). I keep 4 h for checks.

Function	MTD	RTO	RPO	Recovery is declared when	Buffer (MTD − RTO)	Basis
F5	48 h	8 h	8 h	FS01 opens sample records with correct permissions. Records Officer accepts.	40 h	48 h service commitment.
AUTH01 (shared)	n/a	2 h	4 h	Test accounts sign in and groups are correct.	n/a	All five functions depend on it.
DB01 (shared)	n/a	3 h 30 min	1 h	Restore log reaches the chosen point. Consistency check passes.	n/a	F1 to F4 depend on it.

The targets come from the deadlines in the brief. F2 must confirm fees by 16:00 and F3 closes at 18:00, so their MTDs are 8 and 10 hours. F4 has until 12:00 on Day 1, but I cut the MTD to 24 hours so that payroll can be checked before the bank file is sent. F1 has a 12 hour MTD because the examination can run by a workaround while LMS01 is rebuilt. The RPOs follow how fast the data changes. Fees, applications and submissions change every hour, so one hour is the limit. Payroll and records change slowly, so four and eight hours are acceptable.

2.2 Checkpoint analysis

The table shows the age of each checkpoint at 08:00. DB01 can be restored to 07:00, which is one hour old, only if B01 and every incremental to 07:00 are intact. Under A4 that chain is B01 plus eight hourly incrementals. The AUTH01 snapshot is eight hours old and the FS01 backup is nine hours old. Against my RPOs, DB01 just meets one hour. AUTH01 misses its four hour RPO by four hours. FS01 misses its eight hour RPO by one hour.

Item	Checkpoint (WAT)	Age at 08:00	Functions affected	Potential data-loss window	Proposed RPO	Gap against RPO
B02 DB01 chain	07:00 Day 0	1 h	F1 to F4	07:00 to 08:00	1 h (F1 to F3), 4 h (F4)	None. Met, with no margin. The chain must be intact.
B01 DB01 full	23:00 Day −1	9 h	Base for B02	Not a recovery point on its own	n/a	If all incrementals fail, 9 h exceeds the 1 h RPO by 8 h.
B03 AUTH01	00:00 Day 0	8 h	All five	00:00 to 08:00: password resets, new users, group changes	4 h	4 h over (8 h − 4 h)
B04 FS01	23:00 Day −1	9 h	F5	23:00 to 08:00: new or changed files	8 h	1 h over (9 h − 8 h)
Disconnected full	17:00 Day −3	63 h	Emergency use only	17:00 Day −3 to 08:00	n/a	63 h of data loss

Some lost data can be rebuilt from temporary records. Payments made through the external provider can be matched by transaction ID. Applications and learner submissions received through Microsoft 365 are held in mailboxes and forms. Staff notes and tickets can show account changes. Other data cannot be assumed recoverable. This includes anything typed into local systems

between 07:00 and 08:00, new or changed accounts since 00:00, files saved on FS01 since 23:00 and anything held only on flooded local disks.

2.3 Backup and restoration design

The design table closes the RPO gaps. DB01 gets weekly full backups, a nightly differential and hourly incrementals. To restore DB01 I need the latest full backup, the latest differential and every incremental after it, up to the chosen time. For Day 0 that chain is B01 and the eight hourly incrementals from 00:00 to 07:00.

The design follows the 3-2-1 principle. There are three copies, on two media types, with one copy offsite. I add a fourth, independent copy that is immutable and kept offline under a separate administrator identity with multifactor authentication. All copies are encrypted and the keys are held outside AUTH01. Restoration is tested monthly for files and quarterly for the full DB01 chain.

Data set	Type and schedule	Retention	Placement	Controls
DB01	Full weekly (Sunday 23:00). Differential nightly (23:00). Incremental hourly, or every 15 minutes in exam, admission and payroll periods if the platform allows.	Hourly: 7 days. Nightly: 35 days. Weekly full: 12 weeks. Monthly: 12 months.	Copy 1 production. Copy 2 local repository raised above flood level. Copy 3 offsite, encrypted. Copy 4 independent immutable.	Separate backup identity with MFA. AES-256 encryption. Keys outside AUTH01. Alert on every backup admin logon.
AUTH01	Snapshot every 4 hours and after every directory change. Nightly system state export.	Snapshots: 7 days. Nightly: 35 days.	Same four copies.	Same controls. Monthly test restore in isolation.
FS01	Full nightly (23:00). Incremental every 4 hours.	Nightly: 35 days. Monthly: 12 months.	Same four copies.	Same controls. Monthly file restore test.
Independent copy	Immutable copy of the weekly full and nightly differential. Monthly offline full.	Immutable for 90 days. Monthly: 12 months.	Different provider and region, outside the flood zone.	Read only by default. Break-glass credentials in a sealed envelope.
Restoration tests	Monthly file restore. Quarterly timed DB01 chain restore in isolation. Yearly failover drill.	Reports kept 3 years.	Not applicable	Results go to the Principal.

2.4 Why offsite storage is not proof of recovery

Offsite storage alone does not show that data can be restored. A copy can be incomplete, corrupt, encrypted with a lost key or missing an incremental. A live replica copies mistakes and malicious changes within seconds, so it can become a second broken copy.

I use three checks to prove recoverability. For integrity, I verify the catalogue, check hashes and restore the chain in an isolated space. For access independence, I test whether the production administrator identity can read, change or delete backups, and I ask the provider for written controls. For usability, the function owner opens the restored data, compares record counts and checks sample transactions against the source. The unverified assumptions are the hourly

incremental schedule (A4), the provider's access independence (A9) and where the encryption keys are held.

Task 3: Recovery strategy and budget

3.1 Option comparison

Cold cannot meet the targets. Its first critical service takes 24 to 48 hours, which is far beyond the MTDs of 8 to 12 hours for F1 to F3. Hot comes closest to every objective. It also uses the whole NGN 3,000,000 limit and leaves no reserve. A live replica copies corruption and malicious changes, so Hot still needs the independent backups. Warm meets the RTOs on the drill timings and keeps NGN 900,000 in reserve. The quoted windows are planning estimates.

Criterion	Cold	Warm	Hot
First critical service (estimate)	24 to 48 h	3 to 5 h	About 1 h
Data-loss window	Up to 24 h	1 to 6 h	Up to 15 min
Meets RTO (F1 and F2 at 4 h)	No	Probably. Drill gives 3 h 45 min and 3 h 30 min.	Yes
Meets RPO (1 h for F1 to F3)	No	Only with the hourly backup design	Yes
Staffing (two technicians)	Cannot build and restore within the MTDs	Managed in the drill but with no slack	Fewer manual steps. Needs round-the-clock monitoring skills.
Geography	Needs a site and hardware after the event	Warm site must be outside the flood zone and on separate power (A5)	Second site must be well separated. The link must work.
Supplier dependency	High: procurement and supplier availability	Medium: supplier site plus our backups	High: replication link and supplier health
Risk of propagating corruption	Low	Medium. Restore from backups.	High. Replica copies the fault.
Overall	Fails MTD for F1 to F3	Cannot meet the 11:00 exam. RPO depends on our design.	Closest fit but no reserve

Staffing and geography matter as much as cost. All three packages rely on the same two technicians, so a faster package only helps if those two people can run it. The recovery site must also sit outside the flood zone and on a separate power supply. If the site fails for the same reason as the main campus, no package can deliver its quoted times.

3.2 Budget and recommendation

Item	Cold (NGN)	Warm (NGN)	Hot (NGN)
Annual base cost	600,000	1,500,000	2,400,000
Independent protected backups	400,000	400,000	400,000
Training and tests	200,000	200,000	200,000
Complete annual cost	**1,200,000**	**2,100,000**	**3,000,000**

Item	Cold (NGN)	Warm (NGN)	Hot (NGN)
Planning limit	3,000,000	3,000,000	3,000,000
Headroom	1,800,000	900,000	0
Fits the limit?	Yes	Yes	Yes (exactly at the limit)

I recommend Warm at NGN 2,100,000 a year. Its trade-offs are clear. It cannot restore the assessment service before the 11:00 examination, so the continuity workaround in Task 4 is essential. Its data-loss window of 1 to 6 hours meets my one hour RPO only at the best end of the range. I close that gap with the hourly incremental design in Task 2 and I will not rely on the supplier figure until it is tested. The F1 and F2 recovery buffers are only 15 and 30 minutes against the drill.

Residual risk	Why it remains	Accepted by
Assessment service returns about 45 minutes after the exam starts	Warm cannot restore LMS01 before 11:45	Principal with Head of Assessment
Data loss of up to 6 hours if the supplier window applies	Package is weaker than my 1 h RPO unless the hourly design works	Principal with Finance Manager and Registrar
Thin recovery buffers for F1 and F2	The drill has no slack	Principal
Alternate facility may share flood or power risk	A5 is unverified	Principal
Shared administrator identity until fixed	Backups can be reached through production access	IT Manager and Principal

3.3 Comparison with disruption cost and return on investment

Warm would cut the modelled cost of an outage sharply. I price the Warm drill by multiplying each function's elapsed hours by its hourly rate, and I add the NGN 120,000 rescheduling cost to be cautious. If the Microsoft 365 assessment counts as acceptable service (A8), the one-off cost is avoided and the total falls to NGN 505,000.

Function	Elapsed (h)	Rate (NGN/h)	Cost (NGN)
F1	3.75	60,000	225,000
F2	3.5	40,000	140,000
F3	4.25	20,000	85,000
F4	4.25	10,000	42,500
F5	2.5	5,000	12,500
Subtotal			505,000
One-off exam rescheduling (cautious)			120,000
Total with Warm			**625,000**

Outage	No measures (NGN)	Warm, modelled (NGN)	Saving (NGN)
8 hours	1,200,000	625,000	575,000
24 hours	3,360,000	625,000	2,735,000

A single outage cannot establish the annual return on investment. On the 24 hour figures the annual cost of NGN 2,100,000 divided by the saving of NGN 2,735,000 gives about 0.77. Warm pays for itself on this basis only if an outage like this happens about once every 16 months, which is unlikely for a flood. The decision must also rest on how often outages of all kinds occur, how much loss each package prevents, and wider benefits such as learner welfare, data integrity and cyber resilience.

Before approving the spending I need evidence. This includes written supplier recovery times from a test with Northbridge-sized data, service terms with penalties, a flood risk check for the main site and the alternate facility, outage history, an independent timed restore of DB01 and proof that the supplier's access is independent of ours. Until then the supplier claims stay planning estimates.

Task 4: Business continuity and disaster recovery plan

4.1 Scope, activation and coordination

The plan covers the five functions, the main site, local systems, staff and learners, and the 72 hour outage. Activation is triggered when the site is unavailable for more than two hours, when an F1 to F4 deadline is at risk or when AUTH01 or DB01 is lost. The Principal authorises activation. The Deputy Principal acts if the Principal cannot be reached. I recommend activation and run the response. Staff report any trigger to me within 15 minutes and I escalate to the Principal at once.

Four groups work together. The continuity team, led by me, runs temporary services with the function owners. The technical recovery team, led by the IT Manager, restores systems. Facilities reports on site access and safety. The incident response team joins if compromise is suspected and then controls evidence and backup choices. We hold one hourly bridge call and keep one shared decision log. Situation reports go out every hour for the first eight hours and then every four hours. Each report gives the status of F1 to F5, the next estimated time, problems and decisions needed. If email is unsafe we use SMS, phone calls and printed contact cards held by managers. Contact roles are the Principal, the Deputy Principal, the IT Manager, the two technicians, the function owners, the Facilities Manager, the Communications Lead and the account managers at each key supplier. Each role has a named deputy so that no single absence stops the response.

4.2 RACI matrix

R is Responsible, A is Accountable, C is Consulted and I is Informed. Each activity has exactly one A.

Activity	Principal	Coordinator	Tech Lead	Head of Assessment	Other owners	Comms Lead	Facilities	IR Lead
1 Activation of the plan	A	R	C	C	C	I	C	I

Activity	Principal	Coordinator	Tech Lead	Head of Assessment	Other owners	Comms Lead	Facilities	IR Lead
2 Temporary assessment delivery	I	C	C	A/R	I	C	R	I
3 Backup validation	I	C	A/R	I	I	I	I	R
4 Restoration	I	C	A/R	I	I	I	I	C
5 Business acceptance of restored service	A	C	C	R	R	I	I	I
6 Stakeholder communication	C	A	I	C	C	R	I	I
7 Incident response involvement	I	C	C	I	I	I	I	A/R
8 Return to the primary site	A	R	R	C	C	I	R	C

4.3 Temporary procedures for the 72 hour outage

The 25 workstation limit is the main constraint. For the examination, 80 learners use their own devices and 40 need centre workstations. I run 25 centre learners at 11:00 and the other 15 at 13:00 on a different paper version to protect integrity. The Head of Assessment approves this method (A8). Learners with accessibility needs get centre workstations and extra time first.

Function	Temporary procedure	Approvals	Accessibility, integrity and privacy	Records and reconciliation
F1	Own-device learners sit a timed Microsoft 365 assessment at 11:00 with video invigilation. 25 centre learners sit at 11:00 on alternate workstations. 15 sit at 13:00 on a second paper version.	Head of Assessment approves the method. Principal approves the second sitting.	Centre seats and extra time for learners with accessibility needs. Second sitting uses a different paper. Results stored in a restricted folder.	Submissions exported and logged by sitting. After LMS01 returns, marks are uploaded and counts checked against the register.
F2	Learners pay through the provider. Finance staff log each payment in a controlled Microsoft 365 sheet using the provider transaction ID. Fee confirmation at 16:00 only for matched payments.	Finance Manager approves. A second person checks each entry.	No card data stored. Sheet access limited to finance staff.	After FIN01 returns, compare the provider settlement report, the sheet and FIN01. Post missing receipts. Finance Manager signs off.
F3	Applications collected by an online form and a paper form at the centre until 18:00. Each application is numbered. No offers until REG01 or Registrar sign-off.	Registrar approves each batch.	Phone support for applicants without internet. Form results restricted to registry staff.	Enter into REG01 within 24 hours of restoration. Check for duplicates. Count forms against entries.
F4	No payroll run from temporary data. Prepare the payment file from the last approved payroll plus approved changes. Freeze other changes.	HR Manager and Finance Manager both approve before bank submission.	Payroll files encrypted and held only on approved devices.	Compare the file with the last run for duplicates. Reconcile changes in HR01 after restoration.
F5	Log requests in a form. Handle urgent ones by hand after identity checks. Other requests wait for FS01.	Records Officer approves releases.	Check permissions before release. No printing from unapproved copies.	Link every release to the request log. Check files against FS01 after restoration.

The minimum resources are the two technicians plus one supplier technician as a backup, me, five function owners, a communications lead, six invigilators and four finance and registry staff. Equipment includes the 25 alternate workstations, staff laptops, mobile data and secure storage in Microsoft 365. Supplier support is needed from the alternate facility, the payment provider, the bank, the warm site and backup provider, and the mobile network.

4.4 DR runbook

The runbook follows the drill order. Restoring technology is not the same as resuming a business service. A server is technically restored when it starts and passes technical checks. The service has resumed only when the function owner has tested real transactions, checked the data against the 07:00 checkpoint and signed acceptance. Until then the temporary procedures stay in place. If a restored service fails after acceptance, I roll back to the temporary procedure and restore again from the last good point.

Step	Action	Prerequisite	Owner	Completion check	If the step fails
1	Confirm activation. Open the decision log. Call both technicians.	Principal approval	Coordinator	Both technicians on the bridge call	Call the supplier technician and the deputy.
2	Select the backup set: B01 plus incrementals to 07:00, B03 and B04. Check the catalogue.	Offsite catalogue access	Tech Lead	Catalogue times match 23:00, 07:00, 00:00 and 23:00	Use the next older point and tell the Coordinator. Use the independent copy if needed.
3	Build the recovery network (30 min).	Warm site access	Technician 1	Connectivity test passes	Use the supplier standard network image.
4	Restore AUTH01 from B03 (30 min).	Network	Technician 1	Three test accounts sign in. Groups correct.	Rebuild from the latest configuration export. Create key accounts by hand.
5	Security and backup checks (30 min): scan restored AUTH01 and backup files. Check backup admin logons. Verify the chain is complete.	AUTH01	Technician 1 with IR Lead	Scan clean. Chain complete. No unexplained logon.	Stop. Call incident response. Switch to another backup.
6	Restore the DB01 chain (90 min).	Step 5	Technician 1	Log ends at 07:00. Record counts match. Consistency check passes.	Restore to the last good incremental. Record the extra data loss.
7	Restore FS01 from B04 (60 min).	AUTH01 and step 5	Technician 2	Sample files open. Permissions correct.	Restore the previous FS01 backup.
8	Restore LMS01, FIN01, REG01 and HR01 in business order.	DB01	Both technicians	Each application loads and posts a test transaction.	Roll back to the DB01 restore point and retry. Keep the workaround.
9	Check data: sample transactions, row counts and last timestamp against temporary records.	Step 8	Function owners	Owner signs the checklist.	Keep the temporary procedure. Escalate to the Coordinator.
10	Business acceptance. Owner declares the service resumed.	Step 9	Function owners. Principal informed.	Signed acceptance in the log	Continue the manual procedure.
11	Reconcile temporary records into the restored systems.	Step 10	Function owners	Counts match. Second person checks.	Hold new entries and repeat in batches.

Step	Action	Prerequisite	Owner	Completion check	If the step fails
12	Rollback decision if corruption appears after acceptance.	Any time	Coordinator and Tech Lead	Service taken offline. Temporary procedure restarted.	Restore from the independent copy.

4.5 Communication matrix

Stakeholder	Trigger	Message owner	Channel	Approver	Frequency	Alternative channel
Learners	Activation. Exam arrangements. Any change.	Comms Lead	SMS, Microsoft 365 email, website notice	Principal	09:00 then every 2 h on Day 0. Daily after.	SMS and a phone helpline
Staff	Activation. Task changes.	Coordinator	Teams or email, SMS	Principal	Hourly bridge call for 8 h. Then every 4 h.	Phone tree
Principal and governors	Activation. Decisions. Missed deadlines.	Coordinator	Phone then written situation report	Not needed	Hourly then every 4 h	Phone and SMS
Alternate facility	Activation. Extension. Workstation shortfall.	Facilities Manager	Phone and email	Coordinator	On change. Daily.	Phone
Payment provider and bank	F2 and F4 deadlines. Missing payments.	Finance Manager	Provider portal and phone	Principal for payroll	Before 16:00 and before the bank file. On change.	Phone to account managers
Warm site and backup providers	Activation. Backup failure.	Tech Lead	Phone and support portal	Coordinator	Every 2 h during recovery	Phone emergency line
Incident response team	Possible compromise	Tech Lead	Phone then secure message	Coordinator	Immediately then hourly	Phone
Facilities and emergency services	Site safety and access	Facilities Manager	Phone	Principal	As needed	Phone
Media and public	Enquiry or rumour	Principal as spokesperson	Website notice	Principal	On change	Press contact by phone

4.6 Initial learner notice

Subject: Campus closure and your 11:00 examination

Dear learner,

Our main campus is closed today because of flooding and a power failure. We expect it to stay closed for up to 72 hours. Our teams are working to restore our systems.

Your 11:00 examination will still go ahead. If you were told to use your own device, sign in with your Microsoft 365 account at 10:45 and follow the instructions we sent. If you need a campus computer, go to the alternate centre named in your message. Some learners will sit at 11:00 and others at 13:00, so please check your time. Bring your student card.

Please do not use the old campus portal until we say it is working again. We will never ask for your password or payment details by phone or message.

We will send updates by SMS and email every two hours. If you need help, call the helpdesk on +234 800 000 0000 (fictional number).

Thank you for your patience.

Northbridge Skills Institute

4.7 Incident response, return to site and sign-off

Incident response is triggered by a possible compromise. Examples are an unexpected backup administrator logon, a failed integrity check without a clear cause, altered payment records or ransom messages. The Technical Lead tells me within 15 minutes. The incident response lead then controls evidence and backup selection. Restoration from the affected backup stops until it is cleared.

The return to the primary site needs two approvals. Facilities must first clear the site for structure, electrical safety, water damage and fire safety. The function owners must then validate the business service. New records made at the alternate site are moved in controlled batches. Each batch is exported, checked by a second person for duplicates and errors, then imported and counted. The temporary records are kept until reconciliation is complete. The Principal gives the final sign-off after both approvals.

Task 5: Recovery schedule and exercise

5.1 Baseline recovery schedule

The schedule uses two technicians and the current warm site timings. Technician 1 follows the critical path, which is the recovery network, AUTH01, the security and backup checks, DB01 and then LMS01. This path takes 30 + 30 + 30 + 90 + 45 = 225 minutes. So the assessment service is restored at 11:45, which is 45 minutes after the examination starts. This is the earliest possible time. Technician 2 has no work until 09:30 because every other action needs AUTH01 and the checks. All five functions are complete at 12:15.

Action	Technician	Start	Finish	Prerequisite	Minutes
Recovery network	1	08:00	08:30	None	30
AUTH01 restoration	1	08:30	09:00	Recovery network	30
Security and backup checks	1	09:00	09:30	AUTH01 restoration	30
DB01 restoration	1	09:30	11:00	Security and backup checks	90
FS01 restoration	2	09:30	10:30	AUTH01 and security and backup checks	60
LMS01 restoration	1	11:00	11:45	DB01 restoration	45
FIN01 restoration	2	11:00	11:30	DB01 restoration	30
REG01 restoration	1	11:45	12:15	DB01 restoration	30
HR01 restoration	2	11:30	12:15	DB01 restoration	45

The comparison table sets elapsed times against the targets. All five functions meet their RTO and MTD in the baseline. F1 and F2 have small margins. Restoration alone is not enough for F1 because the examination starts before LMS01 is back, so the F1 workaround is required. F2 and F3 meet their deadlines but need interim receipts and applications until 11:30 and 12:15. F4 and F5 need no deadline workaround. On data, DB01 meets the one hour RPO for F1 to F3 and the four hour RPO for F4. F5 misses its RPO by one hour and AUTH01 misses its RPO by four hours.

Function	Finish	Elapsed from 08:00	RTO	RTO met?	MTD	Margin to MTD	Checkpoint age against RPO	Workaround needed?
F1	11:45	3 h 45 min	4 h	Yes (15 min spare)	12 h	8 h 15 min	DB01 1 h against 1 h: met	Yes. The 11:00 exam starts before restoration.
F2	11:30	3 h 30 min	4 h	Yes (30 min spare)	8 h	4 h 30 min	DB01 1 h against 1 h: met	Interim receipts until 11:30. The 16:00 deadline is met after restoration.
F3	12:15	4 h 15 min	5 h	Yes (45 min spare)	10 h	5 h 45 min	DB01 1 h against 1 h: met	Interim applications until 12:15. The 18:00 deadline is met.
F4	12:15	4 h 15 min	6 h	Yes (1 h 45 min spare)	24 h	19 h 45 min	DB01 1 h against 4 h: met	No deadline workaround. Freeze changes until accepted.
F5	10:30	2 h 30 min	8 h	Yes (5 h 30 min spare)	48 h	45 h 30 min	FS01 9 h against 8 h: missed by 1 h	No
AUTH01	09:00	1 h	2 h	Yes	n/a	n/a	8 h against 4 h: missed by 4 h	n/a
DB01	11:00	3 h	3 h 30 min	Yes	n/a	n/a	1 h against 1 h: met	n/a

5.2 Tabletop exercise (simulation)

The tabletop lasts 45 minutes and is a simulation. Its objective is to test decisions when the outage grows, resources shrink, backups fail and records disagree. I play every role in turn. These are the Principal, Coordinator, Technical Lead, Head of Assessment, Finance Manager, Facilities Manager, Communications Lead and Incident Response Lead. The agenda is a 5 minute briefing, then the four injects at minutes 10, 20, 30 and 40 with about 10 minutes each, and a short review. The stress results are kept separate from the baseline schedule.

Inject	Decision	Revised action	Owner	Rationale	Effect on service or recovery
Minute 10: closure extends from 72 hours to five days	Plan for 120 hours of operation from the alternate facility.	Extend the alternate facility booking to at least five days. Ask the supplier for extra technicians so that shifts do not exceed 12 hours. Teach online through Microsoft 365 until LMS01 is accepted.	Principal. Coordinator runs it.	Two technicians cannot work for 120 hours. Learners need teaching to continue.	Recovery times are unchanged. Return to the primary site moves back. Cost rises beyond the 24 hour estimate.

Inject	Decision	Revised action	Owner	Rationale	Effect on service or recovery
Minute 20: alternate facility has only 15 working workstations	Run three centre sittings.	Sittings of 15, 15 and 10 learners at 11:00, 13:00 and 15:00 with a different paper each time. Ask learners who can use a device to sit online. Borrow staff laptops. Ask the facility to repair or add machines.	Head of Assessment. Facilities Manager supports.	The exam can finish on the same day and integrity is kept.	Last sitting ends about 16:00. Invigilation hours rise. The exam is still delivered.
Minute 30: DB01 incremental fails its integrity check. Unexpected backup admin logon.	Stop restoring from this chain. Involve incident response.	IR Lead freezes the backup admin account and rotates credentials from a clean device. Tech Lead finds the failing incremental. Restore B01 and the incrementals before it. If the chain is unusable, use the independent copy or the disconnected backup from Day −3.	IR Lead and Tech Lead. Coordinator informs the Principal.	Possible compromise means the backup cannot be trusted until checked.	DB01 finishes about 30 minutes later at 11:30 and all functions at 12:45. If the 07:00 incremental fails, data is 2 h old, 1 h over RPO. The Day −3 backup would mean 63 h of loss.
Minute 40: provider reports payments missing from restored finance records	Hold fee confirmations and reconcile.	Finance Manager gets the provider's transaction export for 24 hours. Compare IDs with FIN01 and the temporary sheet. Post missing payments with a second person check. If gaps start before 07:00 tell the IR Lead.	Finance Manager	Missing payments within 07:00 to 08:00 match the RPO gap. Earlier gaps point to a restore fault or tampering.	F2 is declared restored with reconciliation pending. 16:00 confirmations go only to matched payments. The technical schedule does not change.

5.3 Corrective actions

ID	Corrective action	Priority	Owner	Target date	Evidence of completion
CA1	Separate the backup administrator identity from production. Add MFA and an alert on every backup admin logon.	Critical	IT Manager	6 Nov 2026	Access review report. Test shows the production admin cannot delete backups. Alert test passes.
CA2	Start hourly DB01 incrementals, 4-hourly AUTH01 snapshots and 4-hourly FS01 incrementals.	High	IT Manager	13 Nov 2026	Schedule screenshots. Catalogue shows entries at the set times for 7 days.
CA3	Run a timed full restore of the DB01 chain in isolation.	High	Tech Lead	27 Nov 2026	Test report. Restore within 90 minutes. Record counts match.
CA4	Confirm the alternate facility has at least 40 workstations and is outside the flood zone. Sign an agreement.	High	Principal and Coordinator	27 Nov 2026	Signed agreement. Flood risk check on file.
CA5	Write and rehearse the payment reconciliation procedure with the provider.	Medium	Finance Manager	20 Nov 2026	Approved procedure. Dry run of 50 payments with no unexplained differences.

ID	Corrective action	Priority	Owner	Target date	Evidence of completion
CA6	Approve the Microsoft 365 assessment contingency with two paper versions.	High	Head of Assessment	20 Nov 2026	Approved procedure. Mock sitting of 25 learners completed.
CA7	Train a third person and a supplier substitute for technical recovery.	Medium	IT Manager	18 Dec 2026	Training list. One drill run with the substitute.

The stress scenario shows that the baseline schedule is optimistic. It assumes every backup passes its checks and every workstation works. When one incremental fails, recovery slows and data loss grows. When workstations are lost, the examination takes longer to deliver. The corrective actions deal with these weak points in order of risk. The shared administrator identity comes first because it threatens every backup. I review progress on all seven actions at each monthly plan review. The Principal closes an action only after seeing the evidence listed.

Task 6: Governance and performance measures

6.1 Ownership, testing and maintenance

The plan owner is the BC/DR Coordinator and the Principal approves it. Each function owner owns the temporary procedure for their function. The IT Manager owns the DR runbook. The plan is reviewed every year and also when something triggers a review. Triggers are a major system or supplier change, a failed test, an incident, a change of key staff, a flood warning for either site and a new audit finding. Role holders are trained within 30 days of starting and refreshed each year.

Every version has an ID, a version number, an approval date and a change log. The current version is stored in Microsoft 365 and also in offline copies held by managers, because the main site may be unavailable. A distribution list records who received each version.

Activity	Frequency	Owner	Notes
Backup job success check	Daily	IT Manager	Report failures the same day.
File restore test	Monthly	Tech Lead	Rotate AUTH01, FS01 and the independent copy.
Timed DB01 chain restore	Quarterly	Tech Lead	In isolation. Compare record counts.
Contact list and call test	Quarterly	Comms Lead	Test SMS and phone tree.
Tabletop exercise	Twice a year	Coordinator	Rotate scenarios: flood, cyberattack, supplier failure.
Warm site failover drill	Yearly	Coordinator	Run before the exam season.
Plan review	Yearly and on triggers	Coordinator	Principal approves.
Training	On joining and yearly	HR Manager	Record attendance.

6.2 Framework mapping and climate consideration

The table maps my outputs to the seven NIST SP 800-34 Revision 1 steps and to ISO 22301:2019 themes (Swanson et al. 2010; ISO 2019). This is an educational alignment only. It does not claim certification.

Output	NIST SP 800-34 Rev. 1 step	ISO 22301:2019 theme
Plan ownership and approval (Task 6)	Step 1: Develop the contingency planning policy statement	Clause 5: Leadership
Business Impact Analysis (Task 1)	Step 2: Conduct the business impact analysis	Clause 8.2: Business impact analysis and risk assessment
Backup design and objectives (Task 2)	Step 3: Identify preventive controls	Clause 8.3: Business continuity strategies and solutions
Option comparison and budget (Task 3)	Step 4: Create contingency strategies	Clause 7.1: Resources and 8.3: Strategies and solutions
Plan, RACI, runbook and notices (Task 4)	Step 5: Develop the contingency plan	Clause 8.4: Plans and procedures and 7.4: Communication
Schedule and tabletop (Task 5)	Step 6: Plan testing, training and exercises	Clause 8.5: Exercise programme

Output	NIST SP 800-34 Rev. 1 step	ISO 22301:2019 theme
KPIs and review triggers (Task 6)	Step 7: Plan maintenance	Clauses 9 and 10: Performance evaluation and improvement

Flooding is a relevant climate issue for Northbridge. ISO 22301:2019 Amendment 1 (2024) adds climate action changes, so organisations should consider whether climate change is a relevant issue for their continuity system (ISO 2024). For Northbridge this means checking the flood risk of the main site and the alternate facility and keeping servers and local backups above likely flood levels. It also means choosing offsite storage in a different area and reviewing power and network routes. A site in the same flood zone would fail in the same way as the main site. Flood warnings are therefore a review trigger.

6.3 KPIs and maturity

KPI	Formula or method	Data source	Target	Frequency	Owner
Recovery time performance	Functions restored within RTO in a test ÷ functions tested × 100	Drill and test logs	100 percent (modelled baseline: 5 of 5)	After each test. Reported quarterly.	BC/DR Coordinator
Data-loss performance	Data sets restored within their RPO ÷ data sets tested × 100	Restore logs	100 percent for F1 to F4	Quarterly	IT Manager
Restoration-test coverage	Backup sets restored and validated ÷ backup sets in scope × 100 (DB01 chain, AUTH01, FS01, independent copy)	Test reports	100 percent each quarter	Quarterly	Tech Lead
Corrective-action closure	Actions closed by target date ÷ actions due × 100	Action log	90 percent or more on time. 100 percent of Critical actions.	Monthly	BC/DR Coordinator
Training	Role holders trained in the last 12 months ÷ all role holders × 100	HR training records	100 percent	Quarterly	HR Manager

References

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Swanson, M., Bowen, P., Phillips, A.W., Gallup, D. and Lynes, D. (2010) Contingency Planning Guide for Federal Information Systems. NIST Special Publication 800-34 Revision 1. Gaithersburg, MD: National Institute of Standards and Technology. Available at: https://doi.org/10.6028/NIST.SP.800-34r1 (Accessed: 23 October 2026).